

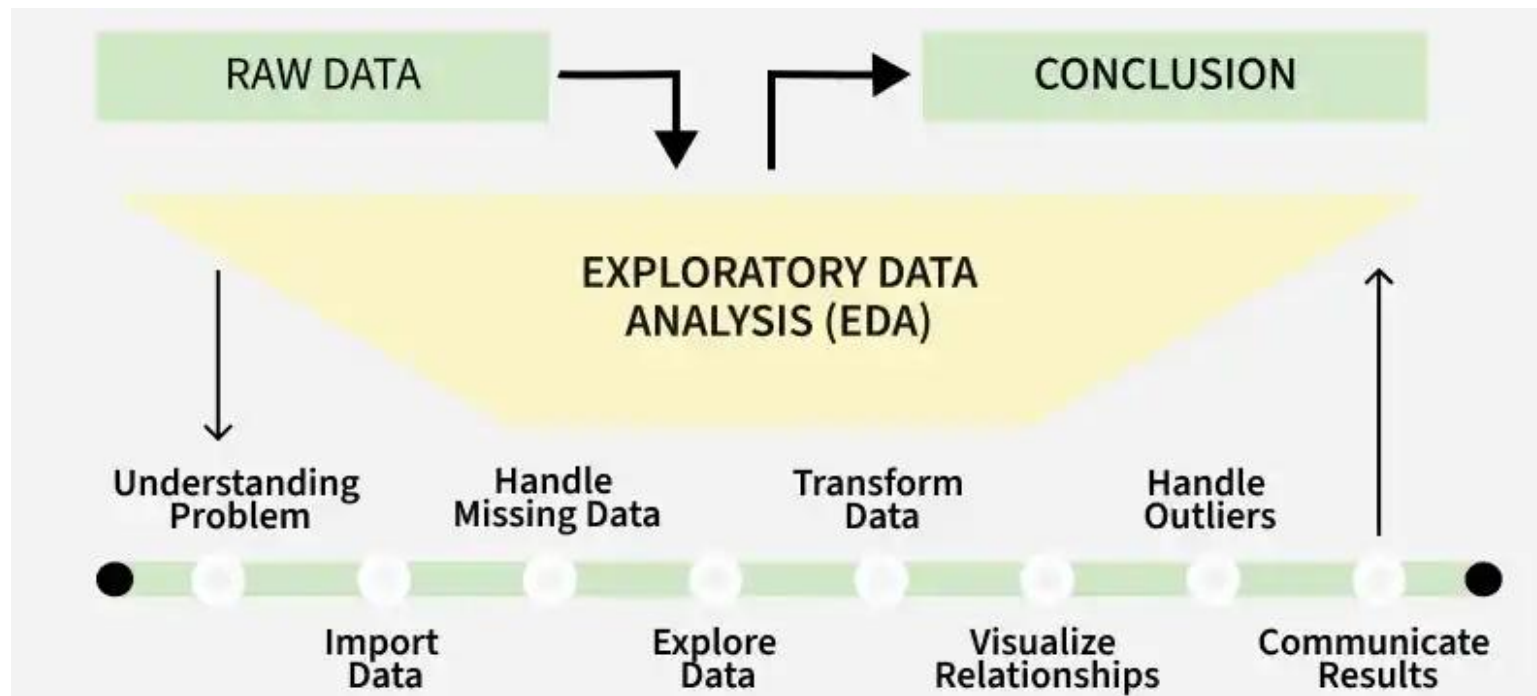
1. Introduction

1.3 Decomposition



EDA

Exploratory Data Analysis (EDA) is the process by which the data scientist analyzes the data by looking at statistics, feature distributions, visualizations, and so on to try and uncover patterns in the data that they can leverage in modeling.



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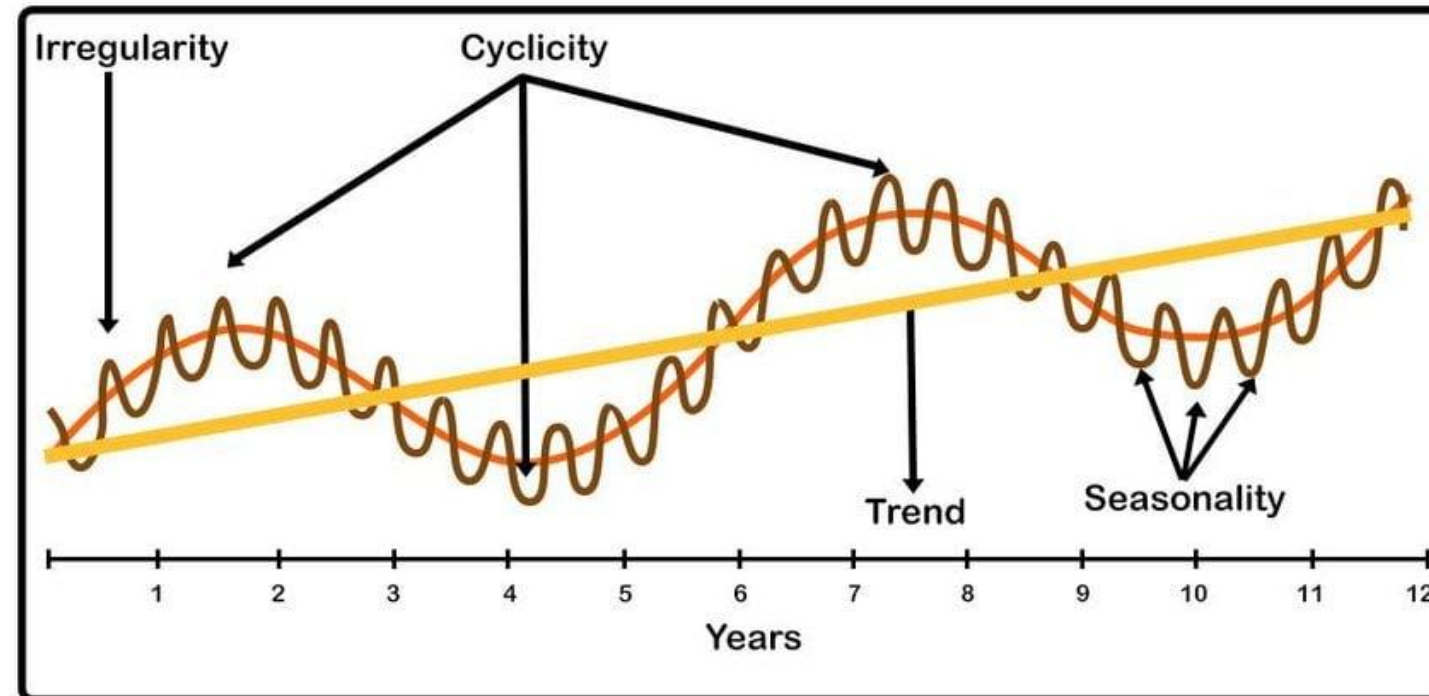
1.3 Decomposition

Components

We can further our understanding of time series by looking at its components:

- Trend
- Seasonal component
- Cyclical
- Residuals (irregular).

In fact, all time series can be decomposed into these elements.



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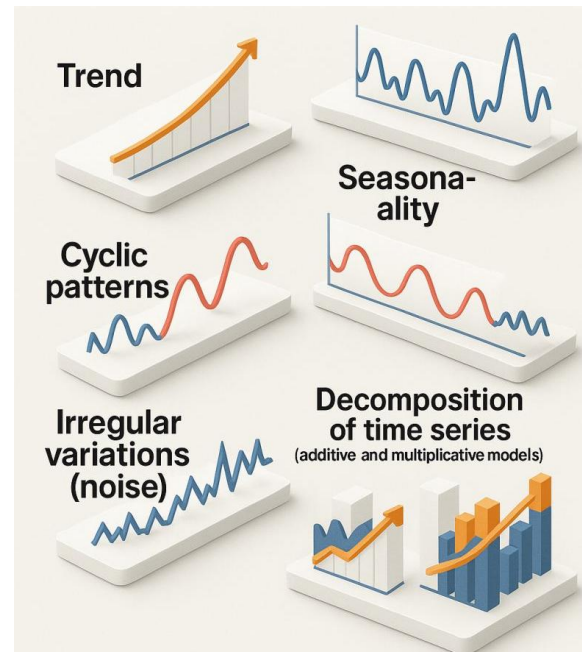
Components

These components can be mixed in different ways, but two very commonly assumed ways are additive

$$Y = \text{Trend} + \text{Seasonal} + \text{Cyclical} + \text{Irregular}$$

or multiplicative

$$Y = \text{Trend} * \text{Seasonal} * \text{Cyclical} * \text{Irregular}$$



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Trend Component

The trend is a long-term change in the mean of a time series. It is the smooth and steady movement of a time series in a particular direction. When the time series moves upward, we say there is an upward or increasing trend, while when it moves downward, we say there is a downward or decreasing trend. The trend doesn't need to be linear; it can also be non-linear.

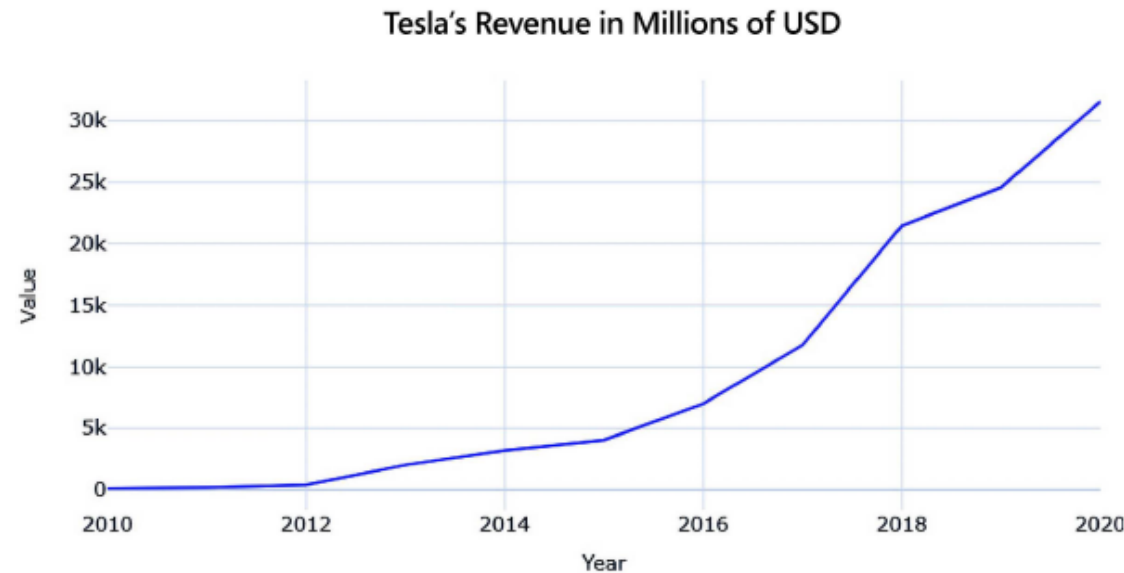


Figure 3.1: Tesla's revenue in millions of USD

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Seasonal Component

When a time series exhibits regular, repetitive, up-and-down fluctuations, we call that seasonality. For instance, retail sales typically shoot up during the holidays, especially during Christmas. Similarly, electricity consumption peaks during the summer months in the tropics and the winter months in colder countries. In all these examples, you can see a specific up-and-down pattern repeating every year.

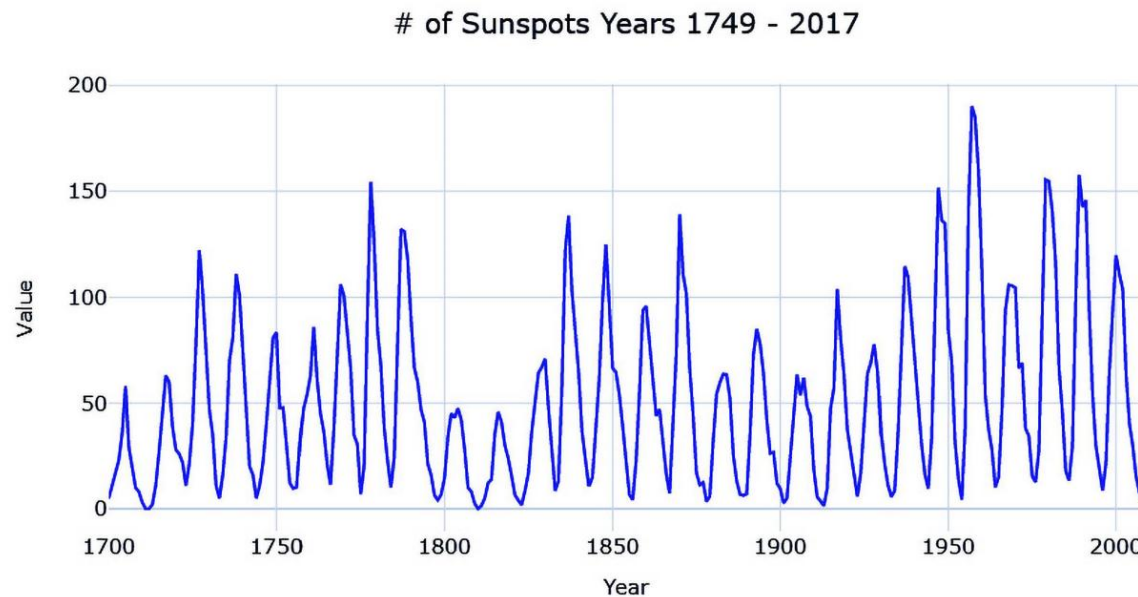


Figure 3.2: Number of sunspots from 1749 to 2017

Sunspots peak every 11 years.

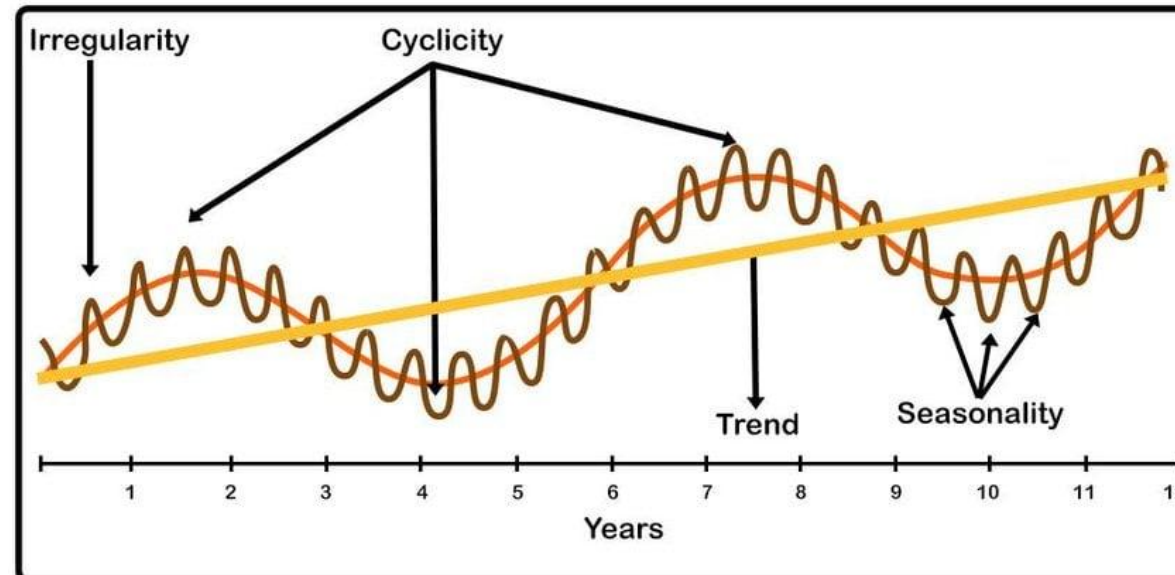
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Cyclical Component

The cyclical component is often confused with seasonality, but it stands apart due to a very small difference. Like seasonality, the cyclical component also exhibits a similar up-and-down pattern around the trend line, but the time over which this cycle moves isn't fixed and is subject to a bit of variation around a general time frame.



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Cyclical Component

- Business activity: GDP, industrial production, or unemployment can rise and fall through expansion, recession, recovery, and renewed expansion.
- Housing market: House prices and construction activity can go through boom and slowdown phases that may last several years, depending on interest rates, credit conditions, and economic confidence.
- Commodity prices: Oil, copper, or agricultural commodity prices may experience multi-year rises and declines caused by changes in global demand, investment, and production capacity.

The common feature is that the pattern rises and falls over time, but the duration of each cycle is not fixed.

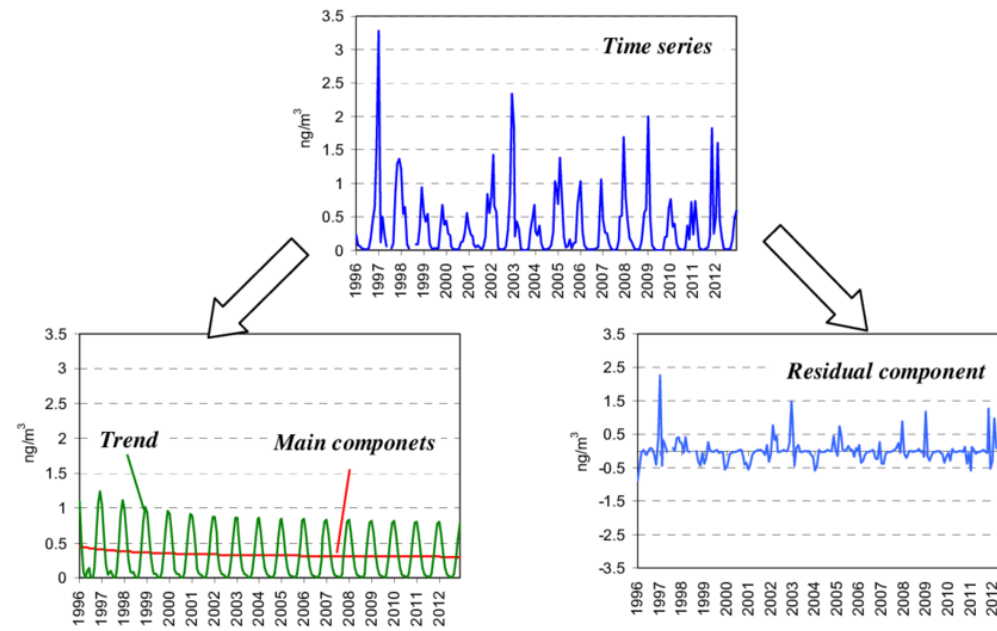
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Residual Component

This component is left after removing the trends, seasonality, and cyclicity from a time series. Traditionally, this component is considered unpredictable and is also called the irregular, error term, or noise term. In common classical statistics-based models, the point of any “model” is to capture all the other components to the point that the only part that is not captured is the irregular component.



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Residual Component

In modern ML, this component is not considered entirely unpredictable. Scientists try to capture parts of this component by using exogenous variables. With additional information, the “unpredictable” component starts to become predictable again.

But no matter how many additional variables you add to the model, there will always be some component, which is the irreducible error, that is left behind. This is the part of the time series which can never be explained no matter how strong the model is or how much additional information you add to the model.